

MACHAGIRI NIRUDU

Software Engineer · Entry Level

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PROFESSIONAL SUMMARY

B.Tech CSE (Data Science) graduate with hands-on experience building full-stack web applications using Python, Django, FastAPI, and React. Built REST APIs with JWT authentication, designed normalized PostgreSQL schemas, and explored AI-powered features using LangChain and RAG pipelines. Comfortable across the stack from database design to frontend UI, with additional exposure to Docker and CI/CD for deployment. Seeking an entry-level Software Engineering / Python Developer role.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS

Frontend: React.js, Vite, Tailwind CSS

Backend: Django, FastAPI, Flask, REST APIs, JWT Auth

AI/Data: LangChain, RAG, Vector Embeddings, Scikit-learn

Databases: PostgreSQL, MySQL

Tools: Git, Postman, Linux, Docker, Jupyter Notebook, OOP, DSA

PROJECTS

Zepto Clone — E-Commerce Platform

[Live Demo](#) | [GitHub](#)

- Built a full-featured product catalog with search/filter, shopping cart, and checkout flow using React.js and Tailwind CSS.
- Developed REST APIs with Django/FastAPI for order processing, inventory management, and JWT-based authentication.
- Designed a normalized PostgreSQL schema for products, orders, and users; deployed the frontend on Vercel.
- Implemented responsive UI components and client-side state management for a smooth cart and checkout experience.

Intrusion Detection System — Major Project

- Built a Python-based machine learning IDS trained on the NSL-KDD dataset, achieving 97%+ detection accuracy.
- Applied Scikit-learn classifiers (Random Forest, SVM) with feature engineering and cross-validation for model tuning.
- Evaluated model performance using precision, recall, and F1-score to minimize false positive alerts.
- Preprocessed and cleaned raw network traffic data, handling class imbalance to improve detection reliability.

DevFlow — Cloud CI/CD Project

[GitHub](#)

- Automated build, test, and deployment pipeline using GitHub Actions, reducing manual release effort.
- Containerized the application with Docker for consistent, portable deployment across environments.
- Configured multi-stage workflows with environment-based secrets management and automated rollback on failure.

EDUCATION

B.Tech – CSE (Data Science) | TKR College of Engineering, Hyderabad | 2023 – 2026

CERTIFICATIONS

- AWS Cloud Practitioner Essentials
- Oracle Cloud Infrastructure AI Foundations
- Software Engineering Job Simulation – JPMorgan Chase & Co. (Forage)